

Toward Understanding the Benefits of Neural Network Parameterizations in Geophysical Inversions: A Study With Neural Fields

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Abstract—Recent research in test-time machine-learning methods has shown that some machine-learning models, without any prior learning, can improve the results of geophysical inversions. Some examples include the deep image prior inversions (DIP-Inv) and the neural fields inversions (NFs-Inv), where the inverse problems are reparametrized by the weights of the machine-learning models. In this work, we employ neural fields (NFs), which use neural networks (NNs) to map a coordinate to the corresponding physical property value at that coordinate, in a test-time learning (TTL) manner. For a TTL method, the weights are learned during the inversion, as compared to traditional approaches, which require a network to be trained using a training dataset. Results for synthetic examples in seismic tomography and direct current resistivity (DCR) inversions are shown first. We then perform a singular value decomposition (SVD) analysis on the Jacobian of the weights of the NN (SVD analysis) for both cases to explore the effects of NNs on the recovered model. The results show that the TTL approach can eliminate unwanted artifacts in the recovered subsurface physical property model caused by the sensitivity of the survey and physics. Therefore, NFs-Inv improves the inversion results compared to the conventional inversion in some cases, such as the recovery of the dip angle or the prediction of the boundaries of the main target. In the SVD analysis, we observe similar patterns in the left-singular vectors as were observed in some diffusion models, trained in a supervised/self-supervised manner, for generative tasks in computer vision. This observation provides evidence that there is an implicit bias, which is inherent in the NN structures, that is useful in supervised/self-supervised learning and TTL models. This implicit bias has the potential to be useful for recovering models in geophysical inversions.

Index Terms—Deep learning (DL), deep neural networks (DNNs), direct current resistivity (DCR), geophysical inversions, implicit bias, inductive bias, neural fields (NFs), seismic tomography, test-time learning (TTL).

I. INTRODUCTION

GEOPHYSICAL inversion is an important step for interpreting observed geophysical measurements [1], [2], [3]. In an inversion, a physical property model of the subsurface is estimated from measurements obtained on the surface, in boreholes or from airborne systems. The quality of the

inversion results depends on robust forward modeling, reliable prior information, and well-designed regularization techniques. Due to the inherently ill-posed nature of geophysical inverse problems, developing a suitable regularization is challenging. Conventional regularization typically penalizes spatial variations and/or explicitly promotes proximity to a reference model [4], [5], but with advances in machine learning, there are opportunities to explore more advanced regularization.

Recently, machine-learning methods have been demonstrated to be successful in many geophysical applications [6], [7], [8], [9]. Alyousuf et al. [7] trained ML models to identify unexpected CO₂ migration. Zhang et al. [8] improved the reconstructions of subsurface seismic velocities by mapping velocity models to raw seismic waveforms using trained conditional adversarial networks. Liu et al. [9] predicted anomalous bodies in the subsurface by matching apparent resistivity data to the resistivity model using trained convolutional neural networks (NNs).

Neural fields (NFs) are advanced ML methods that can solve problems in 3-D rendering, point cloud reconstruction, and biomedical image recovery [10], [11], [12], [13], [14]. NFs use coordinate-based NNs to map spatial, temporal, or other coordinates to the corresponding quantity at those coordinates. NFs are also called coordinate-based signal representations, or neural implicit representations, in certain contexts. Unlike the previously described data-driven methods, which map input-output pairs, NFs use and view NNs as a continuous representation of the target's properties in the coordinate space. Many applications of NFs are not data-driven, instead, they are physics-informed or test-time learning (TTL).

A significant and successful application of NFs is in 3-D rendering, which aims to generate new views of complex scenes using a set of input views [11]. The output values (colors and opacity) of the NFs are fed into a forward map, which is based on a volume rendering algorithm, to generate the predicted observations (pixel values). The objective function, which measures the difference between the predicted measurements and real measurements, is used to update the weights of the NN so that the loss value is expected to decrease iteratively. This training process is very similar to the process of solving inverse problems, so NFs have been adapted to many inverse problems since then. The benefits of utilizing NFs come from the following factors: 1) there is no need for pretraining, namely, the performance of these methods is not limited by the quality and quantity of the training dataset; 2) in

Received 17 March 2025; revised 19 May 2025; accepted 21 June 2025. Date of publication 27 June 2025; date of current version 10 July 2025. This work was supported in part by the Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grants Program and in part by the Mitacs Accelerate Program under Grant IT39167. (Corresponding author: Anran Xu.)

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Digital Object Identifier 10.1109/TGRS.2025.3583970

principle, NFs are mesh-free methods, namely, we can employ an adaptive grid scheme during inversion, which may bring benefits in terms of the accuracy of the reconstruction and the computational costs; 3) parameterizing the inverse problems in a continuous setting naturally introduces smoothing regularization effects; and 4) we can leverage the implicit/inductive regularization/bias from the NN's structure and optimization method.

The performance of NFs methods in 3-D rendering has been shown to be much better than that of conventional methods in terms of distinguishing the boundaries between objects, predicting the pixel values of the main objects [11]. The sources of the benefits of utilizing the NFs are still controversial. For example, Yu et al. [15] argue that the NN is not the source of these benefits by showing that when the NN in the NeRF, a typical NFs method in 3-D rendering, is replaced with spherical harmonics, the performance in the 3-D rendering tasks is not affected. Many other works do not explicitly discuss the sources of benefits by employing the NFs methods in 3-D rendering [11], [16]. A key consideration when assessing and designing a model is its performance in representation tasks. For example, in the context of 3-D rendering, researchers want to find a model that can reconstruct the 2-D image better by learning self-similarity. The main strategies for improving model performance include: selecting a positional encoding function tailored to the problem [17], [18], improving the activation function [19], and employing different bases [15], [20]. Notably, the majority of the works still utilize NNs as a component in the NFs methods.

NFs methods have also garnered significant attention from researchers outside of the domain of computer vision and computer graphics. A main domain for testing the usage of NFs methods is biomedical imaging [13], [21], [22], [23], [24]. In CT and MRI imaging reconstruction tasks, Shen et al. [23] showed that learning the implicit neural representation of the reconstructed images by adapting a prior embedding can significantly improve the resolution of the reconstruction results. Liu et al. [24] used NFs with a pretrained denoising regularizer and showed that this approach could improve the recovery of 3-D refractive index images from discrete measurements. Sun et al. [13] improve reconstruction performance in CT tomographic imaging with a coordinate-based NN by generating new measurements from existing data and incorporating these new measurements during training.

In the geosciences, Rasht-Behesht et al. [25] proposed physics-informed NNs (PINNs) for seismic inversions by parameterizing two physical quantities in the wave equation with two coordinate-based NNs. The results show that this method can provide a good recovery of the subsurface model without the need for an explicit regularization term. Sun et al. [26] tested the performance of parameterizing the seismic velocity directly with a coordinate-based NN in the full-waveform inversion. Both works argue that NFs methods can improve the reconstruction performance in scenarios lacking accurate starting models. NFs have also been applied to the joint inversion of seismic and electrical data. Liu et al. [27] used a Bayesian NN to parameterize the global model properties and a convolutional NN to parameterize the reservoir

properties. The numerical results show that the proposed pipeline provides a more accurate recovery of the reservoir model. Notably, some advanced techniques in computer vision and computer graphics were adapted to some of the scientific inverse problems mentioned above and proved to be useful. For example, both Rasht-Behesht et al. [25] and Sun et al. [26] mentioned the importance of using an adaptive sinusoidal activation function. Liu et al. [24] showed the benefits of adopting a task-specific positional encoding function. There are some works in geophysical inverse problems on parameterizing the subsurface models using NNs [28], [29], but there are only a limited number of works that use NFs methods, and a majority of them focus on seismic inversions. Thus, there is substantial potential to further explore this new area.

There are many previous works on PINNs, which also use coordinate-based NNs in solving scientific inverse problems. In general, PINNs incorporate governing physical laws, such as partial differential equations, directly into deep NNs (DNNs) as prior knowledge. This approach addresses the challenge of data scarcity in training ML models to solve scientific inverse problems [30], [31], [32]. Compared to the direct-feedforward networks, PINNs can achieve better accuracy and shorter training time by using a smaller training dataset [33], [34]. Our work does not fall into the category of PINN mainly because we do not let the NN learn the physics of the problem. Namely, we enforce the PDE constraints by using a PDE solver to calculate the residual instead of using a PDE residual where the PDE is directly embedded in the objective function, so our work do not need any training dataset.

It is well-known that learning alone is insufficient to explain the good performance of deep networks in terms of generalizability. The fact that DNNs can describe the underlying distribution of a dataset, even in high-dimensional spaces that suffer from the curse of dimensionality, suggests that there is a prior (implicit/inductive bias) that has been imposed in the training process [35]. However, the underlying mechanism of this implicit bias has not been fully explained. For example, some researchers have found that certain modern ML models' structures (e.g., Transformers, CNNs, and DNNs) have an implicit regularization or bias that is beneficial for some inverse problems [36], [37], [38], [39]. Some of them suggested that the benefits come from the high dimensionality of the NNs [39], [40], [41]. Others analyze this generalizability using probabilistic theory [42], [43]. Although some of these works have taken a mathematical point of view and have aimed to prove rigorous theorems, they have been limited to overly simplified settings. Other works used the empirical-based method (or the so-called "scientific method"), which forms hypotheses and designs controlled experiments to test them [44]. Specifically, it highlights the importance of empirical analyses of DNNs to validate or refute existing theories and assumptions, as well as to address fundamental questions regarding the mechanisms behind their successes. Although relatively underexplored, these empirical approaches offer significant potential to strengthen our understanding of deep learning (DL) and to drive substantial advancements in both theoretical and practical domains. Some well-known works that emphasize

empirical-based methods include: physics of large language models [44], where Allen-Zhu used synthetic data and built an idealized environment for the training of LLMs to find the “laws of all LLMs.” A scaling law for LLMs states that the performance of LLMs scales with model size, dataset size, and the amount of compute power [45]. Kadkhodaie et al. [35] found that the diffusion models tend to learn a basis that is adapted to geometric features in the underlying images, and they name these bases as “geometry-adaptive harmonic bases (GAHBs).”

Kadkhodaie et al. [35] attributed the implicit bias to the model structure and optimization method, not the training data alone. Therefore, we can utilize the analysis of that work to take one step further to examine the performance of the implicit bias and TTL methods in the geophysical inverse problems. Previous works have demonstrated the benefits of TTL methods to geophysical inversions and have attributed these benefits to the implicit bias provided by the machine-learning model structures [46], [47]. However, these works do not provide an analysis of what priors this implicit bias impose during the inversion. In this work, we will show that test-time machine-learning methods can improve the geophysical inversion result by finding weights that can capture geometric structures in the physical property model.

In this work, we will first introduce our NFs-Inv method and its applications in seismic tomography and direct current resistivity (DCR) inversions. We use the same structure of the NN for both cases. However, we use different positional encoding functions in different cases since some positional encoding functions are helpful in terms of learning high-frequency variations of the subsurface model. The comparisons between the conventional Tikhonov-style inversion results and the NFs-Inv inversion results clearly show that our NFs-Inv method can improve the recovery of the subsurface model in some cases. We will then present our analysis of these inversion results. Our analysis, which uses a singular value decomposition (SVD) analysis on the Jacobian of the weights of the NN, will partly explain why searching over the NN-weights space will improve the inversion results. To the best of our knowledge, this is the first study that applies NFs methods to seismic tomography inversions and DCR inversions and gives evidence to show that the benefits of the TTL method are from the implicit bias inherited from the NN’s structure.

The main contributions of our work include:

- 1) Developing a methodology for using neural fields inversions (NFs-Inv) for geophysical problems in a TTL manner that does not require any pretraining or a training dataset. We also demonstrate the effect/influence of choosing different positional encodings in NFs-Inv.
- 2) Presenting its application on linear/nonlinear geophysical inversions with homogeneous/inhomogeneous background. We compare the NFs-Inv and conventional inversion results. These results demonstrate that the implicit bias from the NFs-Inv is appropriate for some geophysical inversion scenarios.
- 3) Providing an empirically based explanation of the benefits of employing NFs-Inv. The left-singular vectors of the Jacobian have a similar pattern to the “GAHBs”

described by Kadkhodaie et al. [35]. The basis of the Jacobian reveals what has been imposed by the implicit/inductive bias. The bias enables the inversion to overcome the problems of predicting unwanted artifacts due to sensitivity, such as near-electrode artifacts or the concentration along the ray paths. This result suggests that the implicit bias in the NFs-Inv could be similar to the implicit bias found in certain dataset-trained machine learning models, such as diffusion models.

II. METHODOLOGY

Many geophysical inversion methods can be categorized into a physics-driven approach or a data-driven approach. The physics-driven approach imposes explicit regularization to the objective function or uses a problem-specific constrained algorithm during the inversion process. The advantage is that it is a white-boxed approach—namely, the regularization and constraints have well-established geological, mathematical, and/or statistical theories as a foundation. The physics-driven approach can be utilized when there is only limited knowledge of the survey domain, but it can sometimes be very time-consuming since it needs to run a forward simulation in each epoch during the test time. The data-driven approach typically requires more prior knowledge of the geological structures in the subsurface within the survey domain. This prior knowledge can be used to build a dataset of plausible subsurface geophysical models and then can be learned by the ML model trained using this dataset. The trained ML models can give fast predictions for new observations. This approach is black-boxed—namely, it is not guaranteed that the trained ML model can return a plausible subsurface model that fits the observations for any test data. Our work will combine elements of the two approaches: the proposed method is physics-informed since it includes a forward numerical simulation, which will make the method more robust and guarantee that the inversion results can fit the observations to a desired level. The proposed method also directly utilizes the implicit regularization from the coordinate-based representations without the need for a training dataset.

A. Background and Conventional Inversion Approach

This study focuses on cross-hole seismic tomography inversion and 2-D direct current resistivity (DCR) inversion, which are widely used in mining, environmental, and engineering applications [48], [49], [50].

For cross-hole seismic tomography, the goal of the inversion is to recover the subsurface velocity model by analyzing the travel times of seismic waves between boreholes. In this survey, a seismic source is placed in one borehole, and seismic receivers (geophones) are placed in another borehole. The seismic source generates waves that travel through the subsurface and are detected by the receivers in the neighboring borehole. By moving the sources and receivers to various depths within their respective boreholes, multiple arrival times are recorded. The travel times of these waves are then used to create a seismic velocity model of the subsurface, which is related to the lithology and elastic properties of the subsurface.

Using the straight-ray assumption, we can calculate the arrival time at each receiver for a given source using the following equation:

$$t_i = \int_{\text{ray}_i} \frac{1}{v} (x, z) d\ell \quad (1)$$

where t is the first arrival time and v is the velocity of the seismic wave through the subsurface material. ray_i is the i th ray path from the given seismic source to a corresponding receiver.

In a typical DCR survey, transmitters inject a steady-state electrical current into the ground, and receivers measure the resulting distribution of potentials (voltages). The governing equation for the dc resistivity experiment is

$$\nabla \cdot \sigma \nabla \phi = -\nabla \cdot \mathbf{j}_{\text{source}} \quad (2)$$

where σ is the electrical conductivity (which is the reciprocal of electrical resistivity $\rho = 1/\sigma$), ϕ is the electric potential, and $\mathbf{j}_{\text{source}}$ is the source current density. We solve this equation using a finite volume method. For the examples we will show, we use a 2-D tensor mesh. The mesh consists of a core region, with uniform cells, and padding regions on the lateral boundaries and the bottom of the mesh with expanding cells to ensure boundary conditions are satisfied. The forward modeling is implemented using SimPEG [51].

In the conventional method, we use a tensor mesh for both surveys and denote the core domain as $m \in R^{W \times H}$. We initialize $m \in M$ as the reference model m_{ref} , a uniform half-space, and then iteratively update m to reduce the output of the objective function. At each iteration, we input the current subsurface model m into a forward numerical simulation F , whose output $F(m) = d^{\text{pred}}$ is then used to compute the data misfit term

$$\phi_d(m, d^{\text{obs}}) = \frac{1}{2} |W_d(d^{\text{obs}} - d^{\text{pred}})|^2. \quad (3)$$

The objective function is a summation of the data misfit term $\phi_d(m, d^{\text{obs}})$ and the regularization term $\phi_m(m)$ with a trade-off parameter that weights the relative combination of each term. We assume that the noise is Gaussian and depends on the magnitude of the voltage measurements in the DCR survey. A matrix W_d , which includes data uncertainties, has been used to quantify the noise [2]. For the cross-hole seismic tomography survey, we assume that the noise is Gaussian but is independent of the magnitude of the arrival time, so we adapt an identity matrix for W_d .

For the Tikhonov-style geophysical inversions, the regularization term $\phi_m(m)$ generally includes a smallness term and a smoothness term

$$\phi_m(m) = \alpha_s |m - m_{\text{ref}}|^{p_s} + \alpha_x \left| \frac{\partial m}{\partial x} \right|^{p_x} + \alpha_z \left| \frac{\partial m}{\partial z} \right|^{p_z}. \quad (4)$$

Often, the norms applied are $p_{s,x,z} = 2$, however, norms of less than 2 can also be employed to promote sparse or compact structures [4]. Regularization terms are commonly employed for the cases where we have no or limited geological prior knowledge. More advanced regularization can also be employed. For example, Astic and Oldenburg [52] proposed an inversion framework for integrating the petrophysical and

geological prior into the DCR inversion. Laloy et al. [53] and Lopez-Alvis et al. [6] integrated the prior of distribution of the spatial patterns of the subsurface materials into the inverse process using a trained variational autoencoder. In this work, we will perform conventional inversions using the regularization in (4) as the benchmark. The hyperparameter values in (4) can be found in Section III.

B. Proposed NFs-Inv Method

A schematic of the proposed methods is shown in Fig. 1. The key component of the proposed method is to reparametrize the model m by the weights of a multilayer perceptron f . In the 2-D setting, the input of f is the 2-D spatial coordinates of the mesh cell centers (x, y) , and the output is the scalar value of the corresponding physical properties for which we invert (slowness or log-conductivity) on the coordinates.

Positional encoding is a transform function that maps (x, y) to a higher dimensional space before feeding the input vector into the NN to overcome spectral bias. Spectral bias is a well-known experimental finding stating that NNs are biased toward learning lower frequency variations [37]. This is the reason why a direct mapping from the coordinate space to the physical property space using a NN has poor performance in the image representation task—the high-frequency variation has not been learned at the time of convergence, so the reconstructed images are blurred and lack details [37]. Positional encoding was first proposed in the work of NeRF [11] and then explained using the neural tangent kernel theory [17]. For context, we will introduce the positional encodings that are relevant to our work. Let $\mathbf{x} \in R^{w \times 1}$ denote a vector in the coordinate space, and the output of the encoding function is $\gamma(\mathbf{x}) \in R^{h \times 1}$. The common positional encoding functions include [11], [13], [17]

Basic encoding

$$\gamma(\mathbf{x}) = [\cos(2\pi\mathbf{x}), \sin(2\pi\mathbf{x})]^T. \quad (5)$$

Linear positional encoding

$$\gamma(\mathbf{x}) = [\cos(2\pi k^1 \mathbf{x}), \sin(2\pi k^1 \mathbf{x}), \dots, \cos(2\pi k^m \mathbf{x}), \sin(2\pi k^m \mathbf{x})]^T \quad (6)$$

where $k^i = (i/2)$.

Gaussian positional encoding

$$\gamma(\mathbf{x}) = [\cos(2\pi B\mathbf{x}), \sin(2\pi B\mathbf{x})] \quad (7)$$

where $B \in R^{(h/2) \times w}$ is sampled from a normal distribution.

Some other positional encodings, such as the radial Gaussian encoding [24], are not widely used, so we do not conduct experiments on them. The choice of positional encoding functions depends on the problems.

Let us denote this transform function as $\gamma : R^2 \rightarrow R^h$. To obtain the subsurface model from our NN, we require a $W \times H$ number of inputs, which is the same size as the number of active mesh cells in our inversion domain. To simplify the computation, we construct a matrix $Z \in R^{W \times H, h}$ whose rows are the transformed spatial coordinates

$$Z_{ij} = [\gamma(x_i, y_i)]_j \quad (8)$$

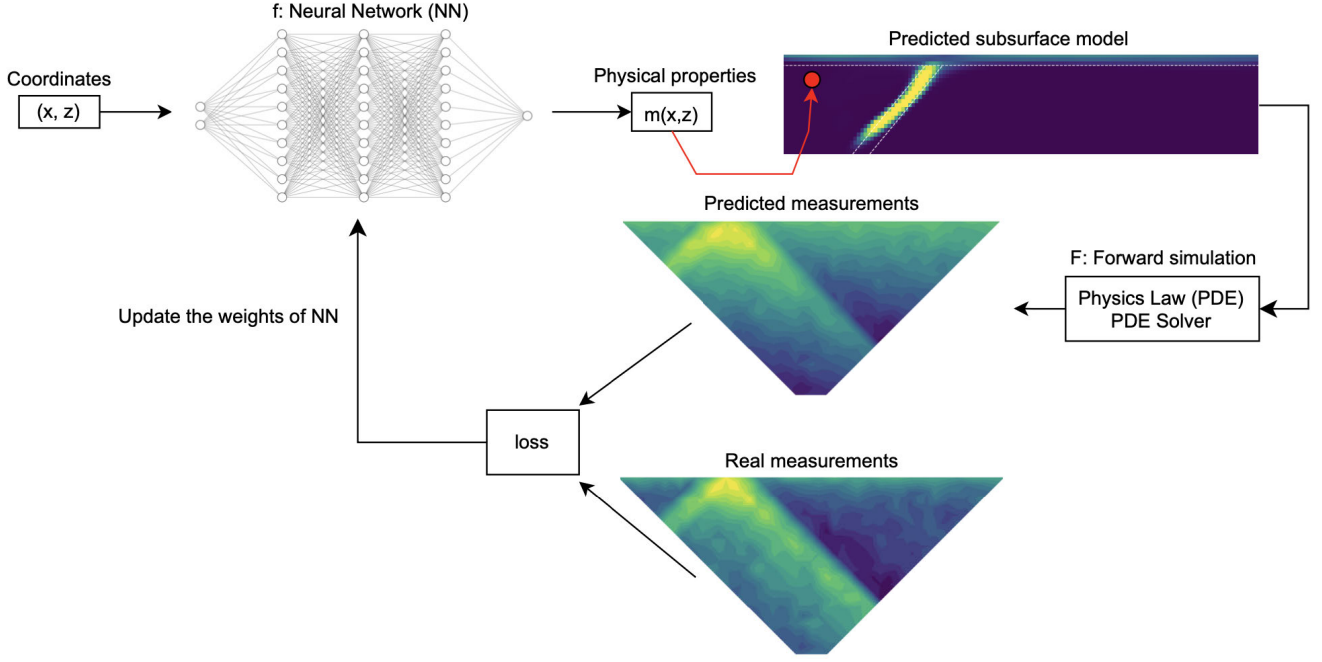


Fig. 1. Proposed inversion pipeline. We reparametrize the subsurface model m by the weights of a NN. Considering a 2-D DCR inversion, the input of NN f is the 2-D spatial coordinates (x, y) , and the output is the scalar value of the corresponding physical property (in this case, the conductivity value) at the coordinates. After obtaining physical properties on all coordinates, we have a 2-D m that is the predicted subsurface model at the current epoch. This 2-D m is then fed into the forward simulation to generate corresponding predicted measurements. After that, the loss function value is computed. During the inverse process, the weights of the NN are updated until the loss value is small enough.

where i is the index for the coordinates and j is the index of the output of the transform function. Then, $W \times H$ can be considered as the batch size. Namely, the inversion results are gained by solving the optimization problem

$$\begin{aligned} \min_m \frac{1}{2} (1 - \beta) \phi_d(m, d^{\text{obs}}) + \beta \phi_m(m) = \\ \min_w \frac{1}{2} (1 - \beta) \phi_d(f_w(Z), d^{\text{obs}}) + \beta \phi_m(f_w(Z)). \end{aligned} \quad (9)$$

There are several strategies for defining the trade-off parameter β . In many cases, a β -cooling strategy is adopted where the inversion starts with a large value for β and β is gradually decreased as the inversion progresses in order to fit the data. In the proposed method, we choose to use the same cooling schedule that is employed by the deep image prior inversion (DIP-Inv) method [47], where a TTL method has been applied to the DCR inversion. The trade-off parameter β in the proposed method for the 2-D DCR inversion is an exponential decay function shown in the following equation:

$$\beta = e^{-\frac{t}{\tau}} \quad (10)$$

where t is the index for the epochs and τ is a constant decay rate. Namely, $\beta = e^{-(t/\tau)}$ in the i th iteration. For the cross-hole seismic tomography inversion, we do not employ any explicit regularization term in the objective function in the proposed method, so $\beta = 0$ for these inversions.

The approach we take for solving this optimization problem is described in Algorithm 1.

Algorithm 1 NFs-Inv Algorithm

Input:

β : the trade-off parameter for DCR inversion,
 Z : encoded input of dimension $R^{W \times H, h}$,
 d^{obs} : observed data (travel time, voltages, etc.),
 m_{ref} : the reference subsurface model,
 f : the multilayer perceptron,

for $t = 1$ to $\dots k$ do

Update the trade-off parameter β^t

Predict subsurface model $m^t = f_{w^t}(Z)$

Calculate the data misfit value $\phi_d(m^t, d^{\text{obs}})$

Calculate the gradient w.r.t. the physical property

$J_v = \nabla_{m^t} \phi_d(m^t, d^{\text{obs}})$

Compute $\mathcal{L}' = (1 - \beta^t)(J_v^T m^t) + \beta^t \phi_m(m^t)$

Update w_t using $\mathcal{L}'$.backward() and optimizer.step()*

end for

Return final surface model m^k

*See DIP-Inv [47] for explanation

C. Architecture of NNs and Optimization

The number of fully connected layers should be adapted to the complexity of the subsurface model and physics (e.g., it is a function of both the resolution of the forward problem and the complexity of the subsurface model). The input dimension of the first hidden layer should depend on the γ function we choose. For example, if we choose the basic encoding (5) in the 2-D setting, then the input should have dimension 4. For both the cross-hole seismic tomography inversion and DCR

inversion examples shown, the results are obtained using a NN, f , with six hidden layers with the leaky rectified linear unit (LeakyReLU) activation function. The first and last hidden layers have 128 neurons, and all other hidden layers have 256 neurons. For the DCR inversions, where we search the solution in the log-conductivity space, the activation function for the last layer is a Sigmoid function. For the cross-hole seismic tomography inversions, the activation function for the last layer is a Tanh function. The output in both cases is multiplied by a constant scalar to fit the possible ranges of slowness/conductivity. The details of choosing the γ function will be discussed in Section III.

Compared to the conventional method, the proposed method searches for a solution in a higher dimensional space, so the number of iterations needed to fit the data is larger. The proposed method employs the Adam optimizer, which is a first-order optimization method. The conventional method in the cross-hole seismic tomography inversion employs a gradient descent optimizer, which is also a first-order optimization method. The conventional method in the DCR inversion employs the inexact Gaussian Newton method, which is a second-order optimization method. The reason why we do not use a first-order optimization method in the DCR inversion is that the forward modeling, which is implemented using SimPEG, is much more time-consuming than the cross-hole seismic tomography inversion, so we want to adapt a second-order optimizer that can take fewer iterations to fit the data. We choose Adam for the proposed method since the adaptive learning rate scheme of Adam is beneficial when updating the weights in the NNs [54]. A second-order optimization method with an adaptive learning rate has been proposed but is limited by the computational cost of estimating the Hessian when the dimension of the parameter is high [55].

III. TRIALS ON THE SYNTHETIC EXAMPLES

In this section, we will consider two geophysical inversion methods: the cross-hole seismic tomography inversion, a linear problem, and the DCR inversion, a nonlinear problem. We present the experiment results on 2-D synthetic models to show the benefits of the proposed method. The proposed method can be readily adapted to 3-D cases by changing the dimension of the input coordinate from two to three. In the cross-hole seismic tomography inversion, the data has sufficient resolution to recover a model even without a regularization, so we do not use an explicit regularization term in the proposed NF-Inv method. We will compare the NFs-Inv result to the conventional method, both with and without an $L2$ smoothness term. In the DCR inversion, the data has less resolution, so in our proposed method, we use an $L1$ smallness term. For the conventional method, we will show results that utilize sparse norms on the smallness and smoothness and include a sensitivity weighting in the objective function.

A. Trials on the Cross-Hole Seismic Tomography Inversions

We will consider two models: the first is a simple model of a block in a uniform background, and the second is an elliptical target in a nonhomogeneous background. For both cases, the

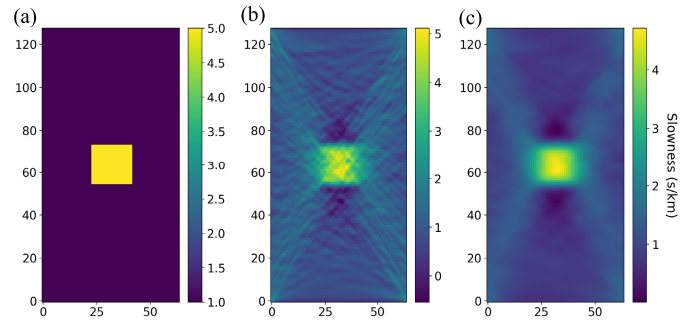


Fig. 2. Inversion results using the conventional method for Case 1. (a) True subsurface model. (b) Conventional result without a smoothness term. (c) Conventional result with the $L2$ smoothness term. The starting/initial model is a uniform half-space for the conventional methods.

tensor mesh has size $W \times H = 64 \times 128$, and each cell has a size of $1 \text{ m} \times 1 \text{ m}$. The two boreholes are separated by 64 m. The receivers are evenly spaced on one borehole with a separation of 1 m. The transmitters are evenly spaced on the other borehole with a separation of 1 m.

In the forward simulation, we calculate arrival times (in units of milliseconds), and these are regarded as our geophysical observations. Gaussian noise with a mean of 0 ms and a standard deviation of 20 ms has been added to the observations for both cases before inversion. The PDE solver block in Fig. 1 refers to the linear forward modeling algorithm.

Fig. 2(a) shows Case 1, a uniform background with a square target. The square target has a velocity of 200 m/s, and the background has a velocity of 1000 m/s. In the conventional method, the initial model is chosen to be a uniform background with a value of 1000 m/s. We perform two conventional inversions. Fig. 2(b) shows the case where no explicit regularization was employed during the inverse process, and the recovered model tends to predict structures along the ray paths. Fig. 2(c) shows the conventional inversion result with only a smoothness term ($\alpha_s = 0$, $\alpha_x = \alpha_z = 0.5$, and $p_x = p_z = 2$). By adapting a smoothness term, the unwanted along-ray artifacts have been eliminated, but we can still see some along-ray-path artifacts remaining.

In the proposed method, the input coordinate space is scaled to $[0, 1]$ in both the x and z dimensions, and the initial model is randomly generated since the $m^1 = f_{w^1}(Z)$ and w^1 is randomly generated using the Kaiming Gaussian initialization method [56]. We choose the basic encoding, so the input dimension of f is 4. For the Adam optimizer, the learning rate is 0.001. In Fig. 3, we show that the proposed method improves the reconstruction by eliminating the unwanted prediction along the ray paths; therefore, the proposed inversion result provides a better recovery of the boundary of the main squared body.

The second case we consider (Case 2) has a heterogeneous background, which was generated using the GeoStatTools package [57], with an elliptical target [Fig. 4(a)]. The conventional method again has the uniform half-space as the initial model.

Similar to Case 1, we performed two inversions. Fig. 4(b) shows the case where no explicit regularization was employed

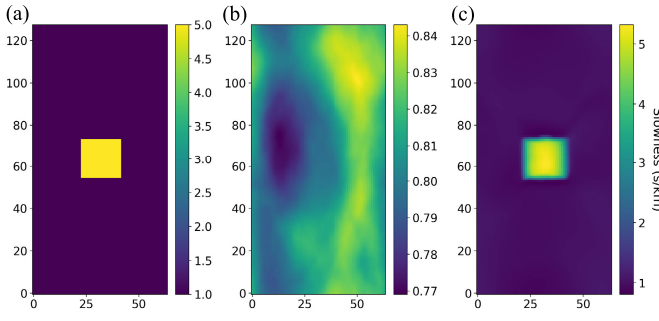


Fig. 3. Inversion results using the proposed NFs-Inv method for Case 1. (a) True subsurface model. (b) Starting model. (c) Recovered subsurface model from the proposed NFs-Inv method without any explicit regularization term.

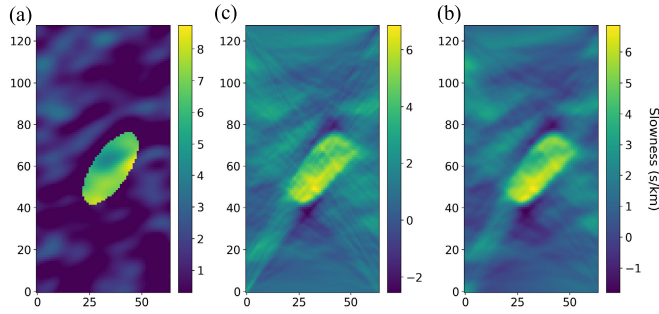


Fig. 4. Inversion results using the conventional method for Case 2. (a) True subsurface model. (b) Conventional result without any regularization. (c) Conventional result with the L_2 smoothness term. The starting/initial model is a uniform half-space for the conventional methods.

during the inverse process, and the recovered model tends to predict structures along the ray paths. Due to the sensitivity and the lack of a positivity constraint, the recovered model has unrealistic negative values in some regions. Fig. 4(c) shows the conventional inversion result with only a smoothness term ($\alpha_s = 0$, $\alpha_x = \alpha_z = 0.5$, and $p_x = p_z = 2$). By adding a smoothness term, the unwanted along-ray artifacts have been eliminated, but the edges of the target are blurred, and we can see some along-ray-path artifacts remaining.

In the proposed method, the input coordinate space is scaled to $[-1, 1]$ in both the x and z dimensions, and the starting model is randomly initialized [Fig. 5(b)]. To have a better resolution on the inhomogeneous background where the high-frequency variations exist, we choose the Gaussian positional encoding with $B \in R^{128 \times 2}$ sampled from a Gaussian distribution with a standard deviation 0.5, so the input dimension of f is 256. The impact of the positional encoding will be discussed in Section IV-B. For the Adam optimizer, the learning rate is 0.001. Compared to the recovered models in the conventional method [Fig. 4(b) and (c)], the NFs-Inv does not have any unwanted along-ray-path structures. The heterogeneous background prediction has more details, and the boundaries of the main target are much better resolved [Fig. 5(c)]. Although we can have a better recovery by imposing a positive constraint (e.g., setting the activation function for the last layer to be ReLU), we employ the hyperbolic tangent as the activation function in the last layer to have a fair comparison with the conventional method where no positive constraint is imposed. Although there is no positive constraint

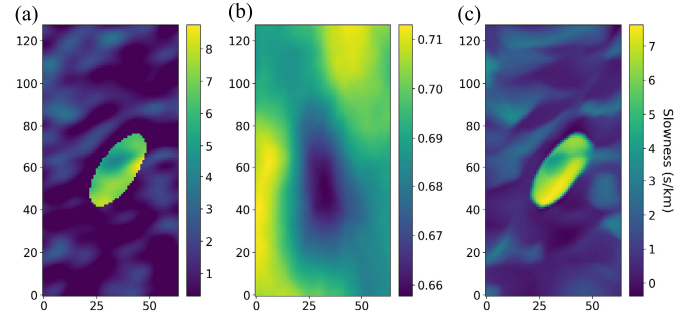


Fig. 5. Inversion results using the proposed NFs-Inv method for Case 2. (a) True subsurface model, (b) starting model, and (c) recovered subsurface model from the proposed NFs-Inv method without any explicit regularization term.

in both methods, the proposed method performs better than the conventional method in terms of having fewer predictions of negative values.

B. Trials on the DCR Inversions

We again consider two cases (Cases 3 and 4). In both cases, there is a dipping target, but in Case 3, the background is homogeneous, whereas in Case 4, the resistivity of the background increases with depth. For both cases, we choose a dipole-dipole survey geometry with a station separation of 25 m and a maximum of 24 receivers per transmitter. The survey line is 700 meters long, so the total number of potential difference measurements is 348.

For the forward simulation, we use a mesh that has core cells of 5×5 m, with 200 core-mesh cells in the x -direction and 45 cells in the z -direction. We add seven padding cells that extend by a factor of 1.5 to the sides and bottom of the mesh. In total, the mesh has $W \times H = 9000$ active cells. In the proposed method, the input coordinate space is scaled to $[-1, 1]$ in both the x and z dimensions in both cases. The input coordinate space only contains the core domain (active cells), and the padding domain values are set to be the background conductivity and frozen during the inversion process. We use the same positional encoding function (identity γ function) and network structure for both cases, and the total number of trainable weights in the NN in f is 263 809.

For Case 3, the top layer has a conductivity of 0.02 S/m. The dike, with three different dip angles, has a conductivity of 0.1 S/m and extends to a depth of 125 m [Fig. 6(a)]. The background has a conductivity of 0.01 S/m.

For the conventional method, we tried a range of parameter choices for the regularization. The best results we obtained used $\beta_s = 1e2$, $p_s = 0$, $p_x = p_z = 1$, $\alpha_s = 0.005$, and $\alpha_x = \alpha_z = 0.5$ [Fig. 6(b)].

For the Adam optimizer, the learning rate is 0.001. For the β -cooling strategy in (10), we choose $\tau = 800$. The results are shown in Fig. 6(d). Considering another TTL method (DIP-Inv [47]) has been developed and tested on the DCR inversion tasks, we also present the inversion result from DIP-Inv for comparison [Fig. 6(c)].

Compared to the conventional methods, both DIP-Inv and the proposed method improve the dip angle predictions.

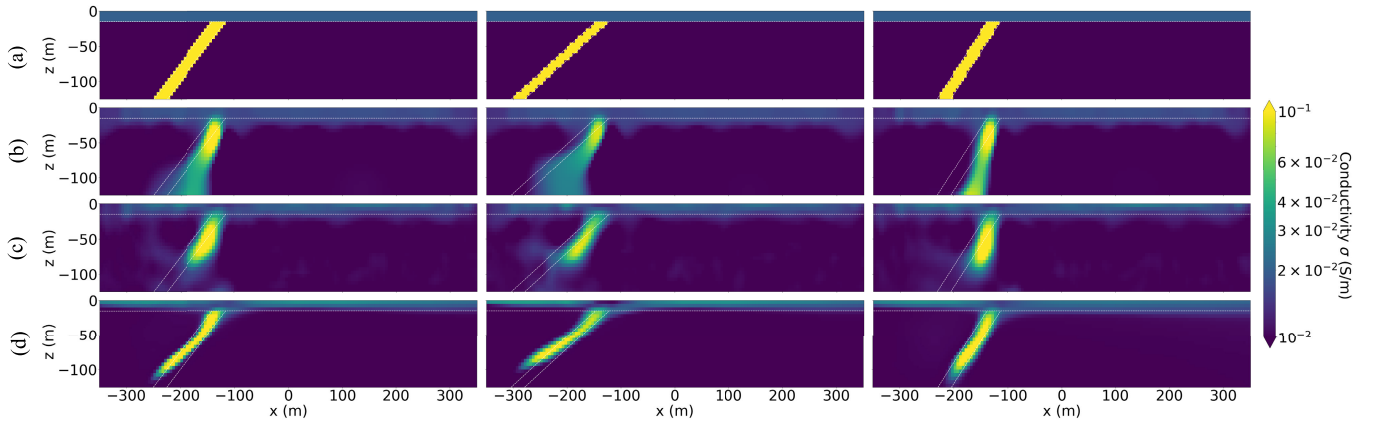


Fig. 6. Inversion results using the proposed method for Case 3. The top row (a) true model. The models in row (b) from the conventional sparse-norm inversions with sensitivity weighting. The models in row (c) from the DIP-Inv. The models in row (d) from the proposed method.

Compared to the results of the DIP-Inv method, the predicted dike structures have sharper boundaries and thinner structures.

To further test the proposed method in a more complicated scenario, we conduct an inversion with the proposed method in a synthetic case with a heterogeneous background [Fig. 7(a)]. The heterogeneous background has a conductivity value that decreases linearly from 0.02 to 0.001 S/m with depth.

Since we assume that we do not have knowledge about the heterogeneous background, we don't have a reference model for Case 4. For the conventional inversion, the smallness term is not applied due to the lack of the reference model, but the smoothness term with $p_x = p_z = 2$ is applied in the inversion process. The starting model is a uniform half-space with a conductivity value of 0.001 S/m. The recovered model from the conventional method can be found in Fig. 7(b).

The recovered model from the proposed method can be found in Fig. 7(c). For the Adam optimizer, the learning rate is 0.001. Although the boundary of the dike structure cannot be recovered as compactly as in Case 3, the location of the main dike is correctly predicted. The conductivity values for the heterogeneous background are predicted approximately correctly, except for the region near the dike structure. Compared to the result from the conventional method, the inversion result from the proposed method has a better depth extent of the dike structure.

IV. IMPLICIT BIAS FROM THE NFs

In this section, we will explore the implicit regularization effect observed in the results from the proposed method. In Section IV-A, we illustrate that by searching in a high-dimensional space and leveraging the implicit bias from the NN, we are able to overcome unwanted artifacts caused by the sensitivity. In Section IV-B, we elaborate on the choice of positional encoding in different scenarios.

A. Benefits of Changing the Searching Space

In Section III-A, the results demonstrate that when searching in the space of the NN-weights for a solution, the inversion results are less influenced by sensitivity artifacts. This is especially noticeable when comparing the conventional and

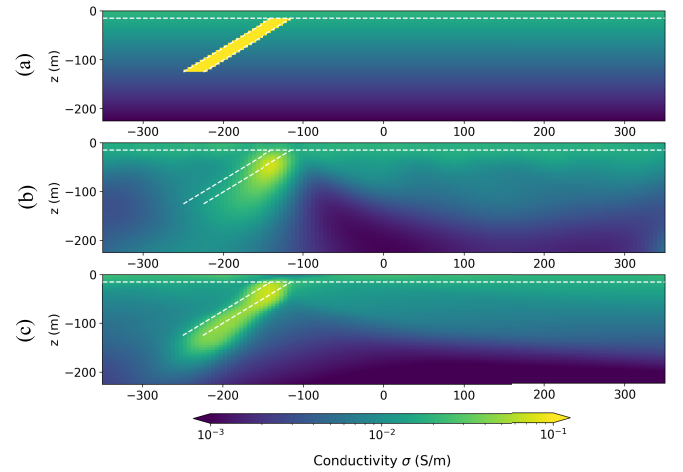


Fig. 7. (a) True model for Case 4. (b) Recovered model from the conventional method for Case 4 (with L_2 smoothness). (c) Recovered model from the NFs-Inv method for Case 4 (without any explicit regularization).

NFs-Inv results when the explicit regularization is not used (Fig. 8). Unlike the seismic inversions, the conventional DCR inversion results are not very meaningful if no regularization is included [Fig. 8(b)]. Namely, searching for the solution in the original mesh space is largely affected by the sensitivity, so the results have unrealistic artifacts near the electrodes [Fig. 8(b)] and therefore the conventional approach cannot correctly predict the structures at depth, such as the dike in Case 3. However, even if we do not apply any regularization to the proposed method (i.e., $\beta = 0$), we are able to recover the main features, for example, the dike and layered structures in Case 3 [Fig. 8(c)]. Considering that the only regularization is from the implicit bias of the proposed method, it is fair to conclude that the NFs methods have a useful implicit regularization effect on the geophysical inverse problem. The challenge of overcoming artifacts due to the sensitivities of a given survey is common across many geophysical experiments (e.g., it is common to see structures cluster around ground electromagnetic stations). This also indicates the potential application of the NFs-Inv method to other geophysical data types.

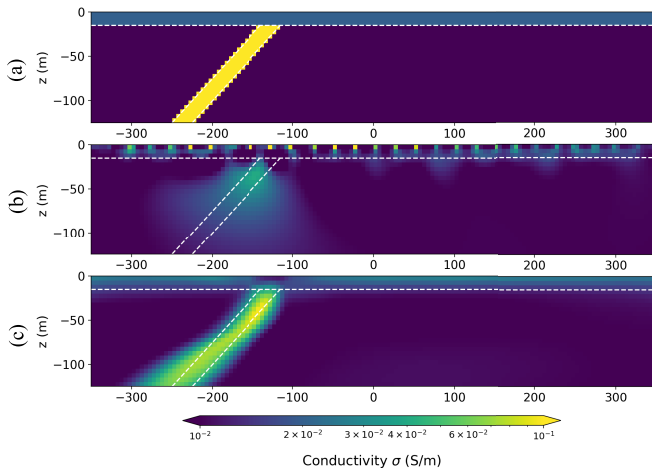


Fig. 8. (a) True model for Case 3. (b) Recovered model from the conventional method without any explicit regularization for Case 3. (c) Recovered model from the proposed NFs-Inv method without any explicit regularization for Case 3.

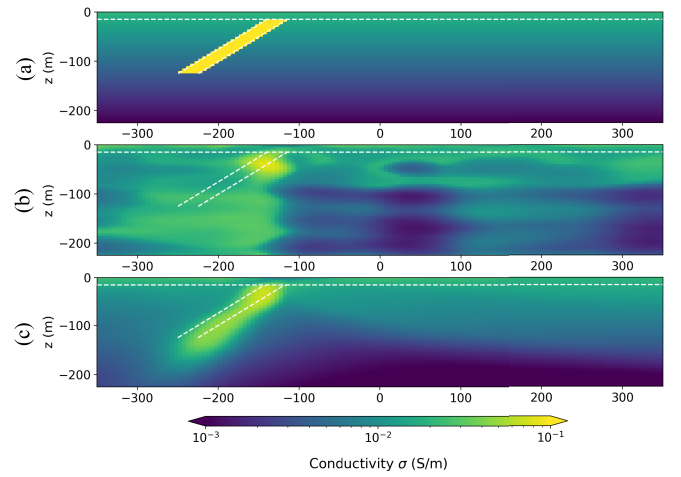


Fig. 10. (a) True model for Case 4. (b) Proposed NFs-Inv result with a linear positional encoding where $m = 8$ in (6). (c) Proposed NFs-Inv result with an identity γ .

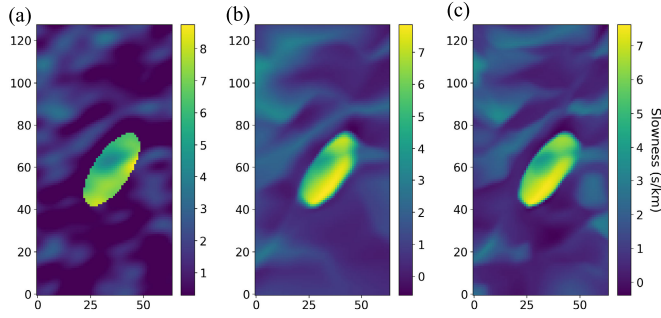


Fig. 9. (a) True model for Case 2. (b) Proposed NFs-Inv inversion result with an identity γ . (c) Proposed NFs-Inv inversion result with a Gaussian positional encoding where $B \in \mathbb{R}^{128 \times 2}$ in (7).

B. Apply Positional Encoding to Recover the Higher Frequency Variation

To further test the usage of the positional encoding (γ), we conduct an ablation study on Cases 2 & 4 where we apply the proposed method with/without positional encoding. We observe that the choice of positional encoding is problem-dependent. For the results presented in this section, we use a Gaussian positional encoding for Case 2 (seismic tomography) and a linear positional encoding for Case 4 (DCR).

From Fig. 9, we observe that without a positional encoding, the recovered model is smoother and details in the background are lost [Fig. 9(b)]. By using a positional encoding [Fig. 9(c)], we recover a sharper edge to the boundary of the main target and find details in the background slowness model. In contrast, adding positional encoding will give unwanted disturbances to the prediction of the main structures in Case 4 [Fig. 10]. Notice that the inversion results in Fig. 10 fit the observations at the same level, but the recovered models differ a lot. This evidence shows that the γ function also introduces a bias to the geophysical inverse problems.

As mentioned before, mapping the inputs to a higher dimensional space using high-frequency functions prior to passing

them to the network enables better fitting of high-frequency variation data. That is why we observe a better recovery of the heterogeneous object in the seismic tomographic example. The fact that we observe the deteriorated recovery of the model in Case 4 illustrates that the choice of positional encoding should align with the ability of the geophysical survey to resolve the subsurface model. Namely, the positional encoding introduces an additional bias to the geophysical inverse problems, which may deteriorate the results in some cases. The resolution of a survey depends upon the underlying physics of the problem, survey geometry, and geologic complexity. In the case of cross-hole seismic tomography inversion, the sensors are located on both sides of the survey domain, so there are enough data to constrain the possible solutions. The inversion results with or without the positional encoding can recover similar contours of the subsurface model. Therefore, adding a positional encoding can be helpful for recovering the high-frequency component in the subsurface model in the seismic case. In contrast, the DCR survey for Case 3 & 4 only has sensors on the ground, and the survey domain is larger than Case 1 & 2. As a result, Case 3 & 4 is much more ill-posed than Cases 1 & 2. Namely, the solution is less constrained, and there are more geologically plausible solutions that can fit the observations to the same degree. In the DCR examples, the additional bias provided by the positional encoding may cause the inversion result to have unrealistically high-frequency signals. As a result, whether we choose to employ a positional encoding should depend on the type of survey method. In terms of selecting the optimal positional encoding, we find that the Gaussian positional encoding with a small standard deviation for generating the matrix B [see (7)] can produce similar results as the basic positional encoding. The selection of the hyperparameters of the positional encoding depends on the complexity of the subsurface model. A more practical guide for choosing positional encodings is an area for future research and will be informed by conducting more trials on different subsurface models and survey methods.

V. ANALYSIS USING THE SVD ANALYSIS OF THE JACOBIAN OF THE NN

In this section, we perform an empirical analysis to examine the implicit bias that is utilized in the NFs-Inv approach. We hypothesize that this bias is similar to the implicit bias that is utilized in the generative models and which helps generalizability.

While DL has demonstrated remarkable success across a growing range of tasks, a comprehensive understanding of the underlying principles driving these achievements remains limited. Many empirical-based methods (“scientific method”) have been developed for studying the underlying mechanism [35], [44], [45]. For example, Kadkhodaie et al. [35] found that deep generative models have inductive biases toward learning a geometry adaptive harmonic basis, and it could potentially lead to solving the generalization puzzle in many large ML models. They acknowledge that there must be priors from the NN structures and the optimization methods and refer to it as inductive bias. For clarification, we use implicit bias instead of inductive bias in our context to avoid potential ambiguity since some papers refer to inductive bias in CNN specifically as translational invariance [58]. Kadkhodaie et al. [35] stated that since a DNN can be considered as a locally linear function, conducting an eigendecomposition analysis (SVD analysis) of the Jacobian of the NN with respect to the input can reveal what has been learned by the generative models. Namely, since these left-singular vectors form the basis of the output, the features of the basis can indicate what has been imposed by the implicit bias during the training process. They find that the learned bases have oscillating harmonic structures along contours and in homogeneous regions. They name these bases as GAHBs.

In our work, we also find that the Jacobian has a geometry-adaptive harmonic basis after inversion. Note that Kadkhodaie et al. [35] conducted a SVD analysis of the Jacobian of the output with respect to the input. In their setting, the inputs of the model are different for different input images. However, in our case, since the inputs are always the fixed values of the coordinates of each cell, we conduct a SVD analysis of the Jacobian of the output with respect to the weights of the NN f . Since the choice of the positional encoding may introduce an additional bias (See Section IV-B), we do not employ positional encoding for the models presented in this section (i.e., γ is an identity function) to avoid potential ambiguity. The SVD of the Jacobian is

$$J = \frac{\partial m}{\partial w} = U \Lambda V^T. \quad (11)$$

The left-singular vectors (the columns of U) have the dimensionality of the geophysical model, and the right-singular vectors (the columns of V) have the dimensionality of the number of weights in the NN. Λ is a diagonal matrix whose elements are the singular values λ_i . We focus our analysis on the left-singular vectors, as these can be thought of as the learned basis for our geophysical model.

Fig. 11 shows the first ten left-singular vectors for Case 1. As observed in Kadkhodaie et al. [35], the first few left-singular vectors strongly highlight the edges of the main

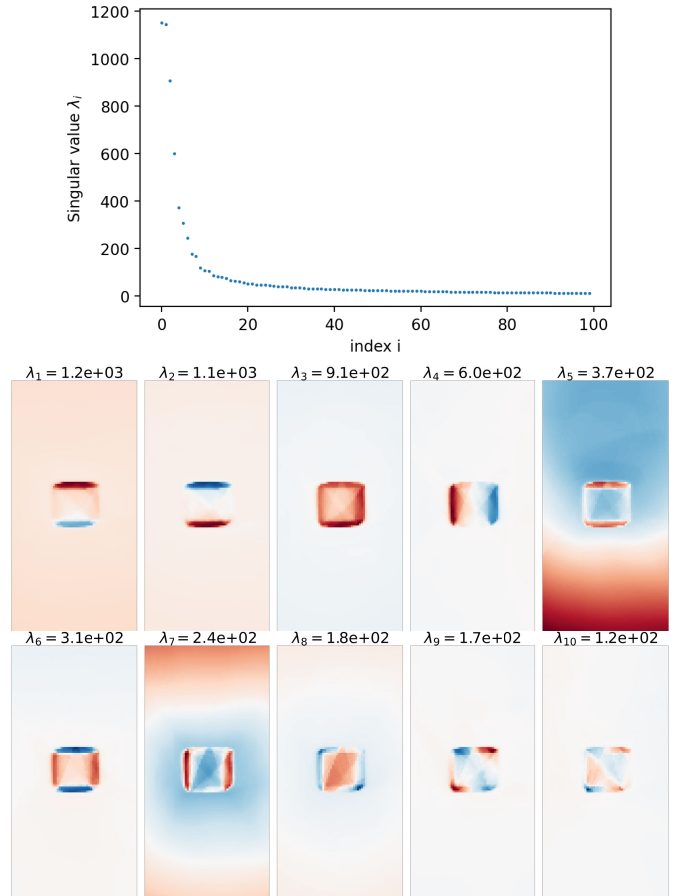


Fig. 11. First 100 singular values and first ten left-singular vectors for the Jacobian in Case 1.

target, while the background is relatively homogeneous (or smooth in cases where there is a structure). Fig. 12 shows the first ten left-singular vectors for Case 2. We can observe that the left-singular vectors capture the main geometric structures related to the elliptical target. The first three left-singular vectors highlight the main elliptical target and are relatively homogeneous in the background. Starting from the fourth left-singular vector, we then see more structures associated with details and higher frequency variation in the background.

Figs. 13 and 14 show the first ten left-singular vectors for the DCR inversion cases. Similarly, we observe harmonic features along the contour of the main dike target in the DCR case. The first two left-singular vectors highlight the dike structure. Starting from the third left-singular vector, we also see the influence of the near-surface layer.

From the plots of the eigenvalues for all four examples, we find that the eigenvalues are sparse. Namely, the first few eigenvalues are significantly larger than the remaining eigenvalues. In Case 1 and Case 3, which are both relatively simple, there is a reduction of nearly a factor of 10 between the first and tenth singular values. For Case 2, which is more complex, there is a reduction of a factor of around 3 between the first and tenth singular values. This observation agrees with the finding in the analysis of Jacobian in the diffusion models [35], where they also find this shrinkage of the eigenvalues.

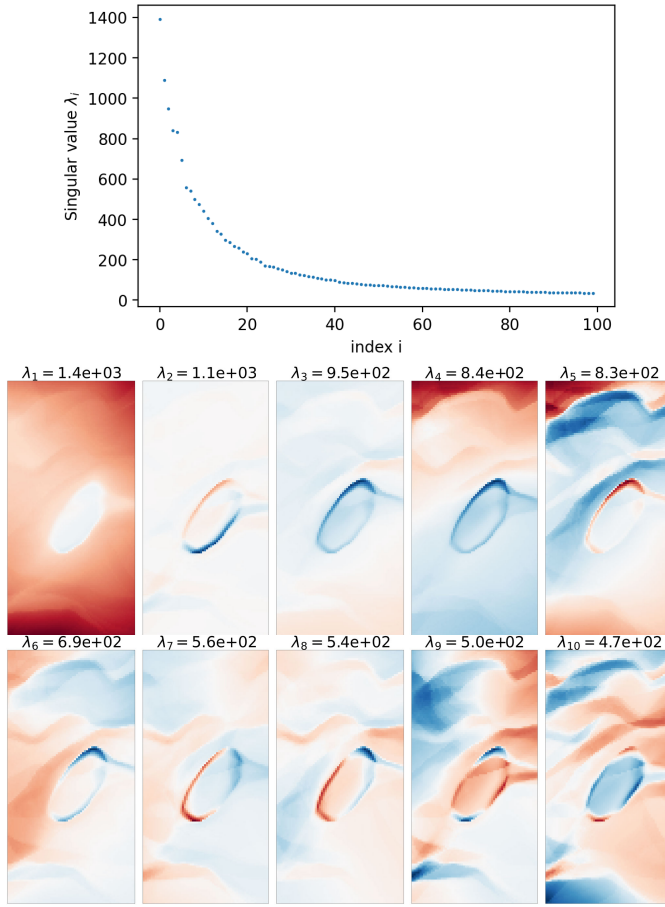


Fig. 12. First 100 singular values and first ten left-singular vectors for the Jacobian in Case 2.

These examples are evidence that there exists an implicit bias, which has been illustrated in the self-supervised learning models and may also be utilized in TTL models as well. Moreover, this analysis can partly explain the reason why we can observe a better recovery of the subsurface physical properties when we reparametrize the subsurface model using the weights of a DNN. By forming these adaptive bases, we have the chance to overcome the problems of predicting unwanted artifacts due to sensitivity, such as near-electrode artifacts or the concentration along the ray paths.

VI. DISCUSSION AND FUTURE WORK

Further work, such as the applications to the field data and other theoretical analyses of the implicit bias, is still in progress. It has been shown that using the adaptive grids can improve the performance of the NFs methods [111], [59], so testing the adaptive grid scheme for geophysical inversions is a possible future research direction.

The run-time needed for the NFs-Inv algorithm heavily depends on the time required to run the forward simulation. For the seismic tomography examples, since the problem is linear and can run on a GPU, the computational costs for both the conventional and proposed methods are small and similar. For the DCR inversion, the runtime between the conventional and the proposed method differs significantly. The computational

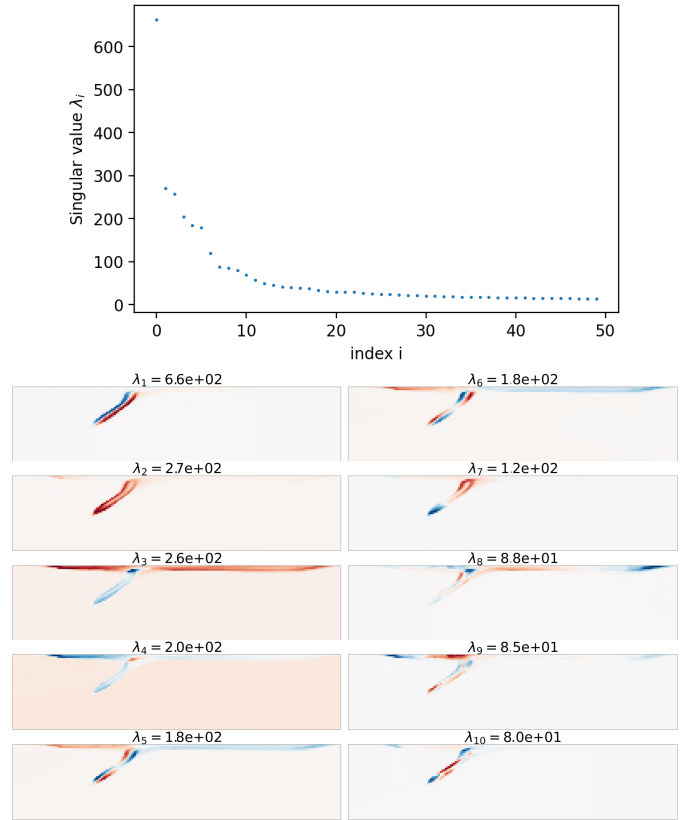


Fig. 13. First 50 singular values and first ten left-singular vectors for the Jacobian in Case 3.

time for the conventional method using sparse-norm inversion with 44 epochs is 466 s. For our proposed approach, we ran for 2000 epochs; however, in practice, there is a minimal difference between the solution observed at 1000 epochs and the final solution. The computational time for the proposed method with 1000 epochs is 9708 s. The CPU time is averaged over four trials running on a 48-core CPU server with Intel(R) Xeon(R) CPU E5-2670 v3 @ 2.30 GHz. The time per epoch is similar for both approaches, as 98.5% of the time is spent on the forward mapping and calculation of the Jacobian matrix using SimPEG. Future work on improving the computational cost includes designing a suitable optimization to decrease the number of epochs needed and improving the efficiency of the forward simulation.

The results shown in Section III are state-of-the-art, which we establish by testing the performance of different positional encoding functions. We find that in the cases where the subsurface model is relatively simple, such as Cases 1 & 3, adding a positional encoding function where h is large will deteriorate the inversion results. However, the results of Case 2 show that when the model is more complex and we use a geophysical method with sufficient sensitivity, adding positional encoding can help us to recover sharper boundaries and finer details in the model. There is still some work left on exploring the effect of the choice of the γ function on the proposed method.

We have tested employing a sinusoidal function [19] as the activation function in Cases 2 & 4, and the results are not ideal since it decreased the sharpness of the boundaries of

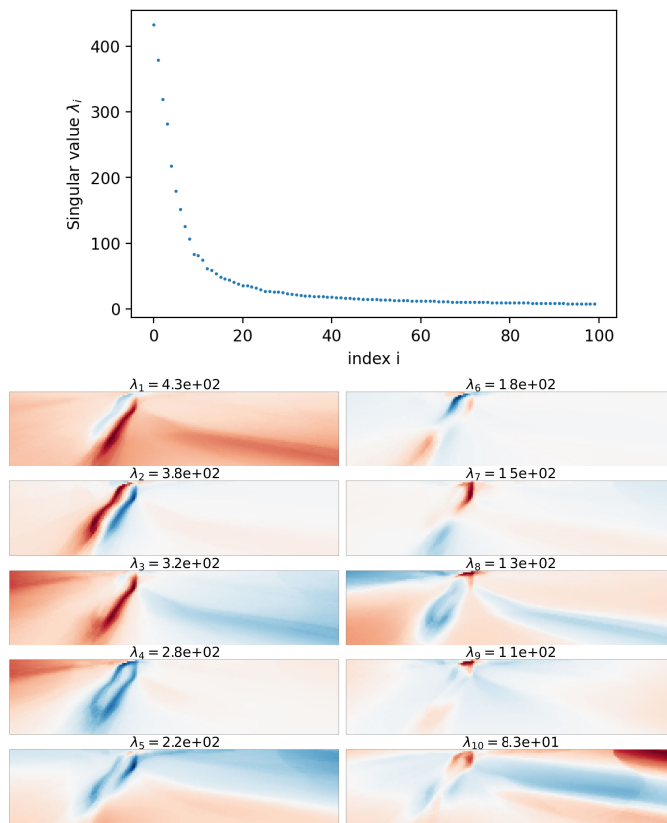


Fig. 14. First 100 singular values and first ten left-singular vectors for the Jacobian in Case 4.

the main target. Kadkhodaie et al. [35] found that ReLUs are key for observing geometric adaptivity in the SVD analysis on diffusion models. We hypothesize that the LeakyReLU activation layer is potentially a critical component of the implicit bias we observed. More tests on the performance of the adaptive activation function should be done in the future.

VII. CONCLUSION

In conclusion, the application of NFs in geophysical inverse problems, as explored in this work, demonstrates significant advantages over conventional inversion methods. By framing the inverse problem in terms of the NN weights, we leverage the implicit bias of NFs. Using a coordinate-based network allows us to take advantage of the built-in smoothing effects that come from parameterizing the inverse problem in a continuous setting. The results we showed, using seismic tomography and DCR, highlight the ability of NFs to reduce unwanted artifacts and improve the accuracy of subsurface models, particularly in capturing the boundaries and dip angles of key targets. This TTL approach proves to be a promising direction for enhancing inversion results in geophysical problems.

Moreover, using an empirical approach, we provide evidence that there exists an implicit bias, which has been illustrated in self-supervised learning, that can also be utilized in TTL models. By forming an adaptive basis during the inversion, we have the chance to overcome the problems of predicting unwanted artifacts due to sensitivity, such as

near-electrode artifacts (DCR) or the concentration along the ray paths (seismic). This analysis can partly explain the reason why we can observe a better recovery of the subsurface physical properties when we reparametrize the subsurface model using the weights of a DNN. By showing that the implicit bias brought by the DNNs can benefit geophysical inversions, we also give insights into other machine-learning methods in geophysical problems.

DATA AVAILABILITY

The code used is available at <https://doi.org/10.5281/zenodo.15034077>

ACKNOWLEDGMENT

The authors would like to thank Zahra Kadkhodaie for elaboration of her works. They also want to thank UBC GIF group members for their invaluable guidance and advice on the usage of SimPEG. They are grateful for the careful review and constructive feedback provided by the reviewers.

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